

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-EA-5601-06 Sheet 1 of 1
OPL Title	Fouling Trend & Cleaning Schedule (PDT-5605)	Discipline	Instrument	
Equipment	EA-5601 - SOLVENT HEATER	Area / Unit	5600 - SOLVENT HEATING SYSTEM	
Related Interlock	N/A (control loop only) (N/A)	P&ID Ref	TJC-LLD-PID-5601	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D- S -M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **fouling trend & cleaning schedule (pdt-5605)** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. Trending the tube-side differential pressure predicts when cleaning is due before the duty is lost. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Log PDT-5605 differential pressure weekly	Loop isolated / under permit
2	Note the PDAH-5605 fouling alarm at 0.7 bar dP	Reference standard traceable
3	Correlate rising dP with any outlet-temperature shortfall	As-found & as-left recorded
4	Plan cleaning before the duty loss becomes critical	Trip / control action verified
5	Feed the trend into the exchanger cleaning program	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Outlet temperature hunting around set-point	Steam valve TV-5602 loop poorly tuned	Retuned TIC-5602 PID, verified stable control
Erratic tube dP reading	Impulse lines plugged with fines	Cleaned and re-zeroed PDT-5605 transmitter
Cracked gauge glass on condensate pot	Thermal/mechanical stress on glass	Replaced LG-5606 gauge glass and seals

6. KEY LEARNING POINTS

- Trending the tube-side differential pressure predicts when cleaning is due before the duty is lost.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Monday, 25 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.